# Sc2207: Intro to Databases
# Lab 3 Assignment
# Lab Group: SCSC

TEAM 1:
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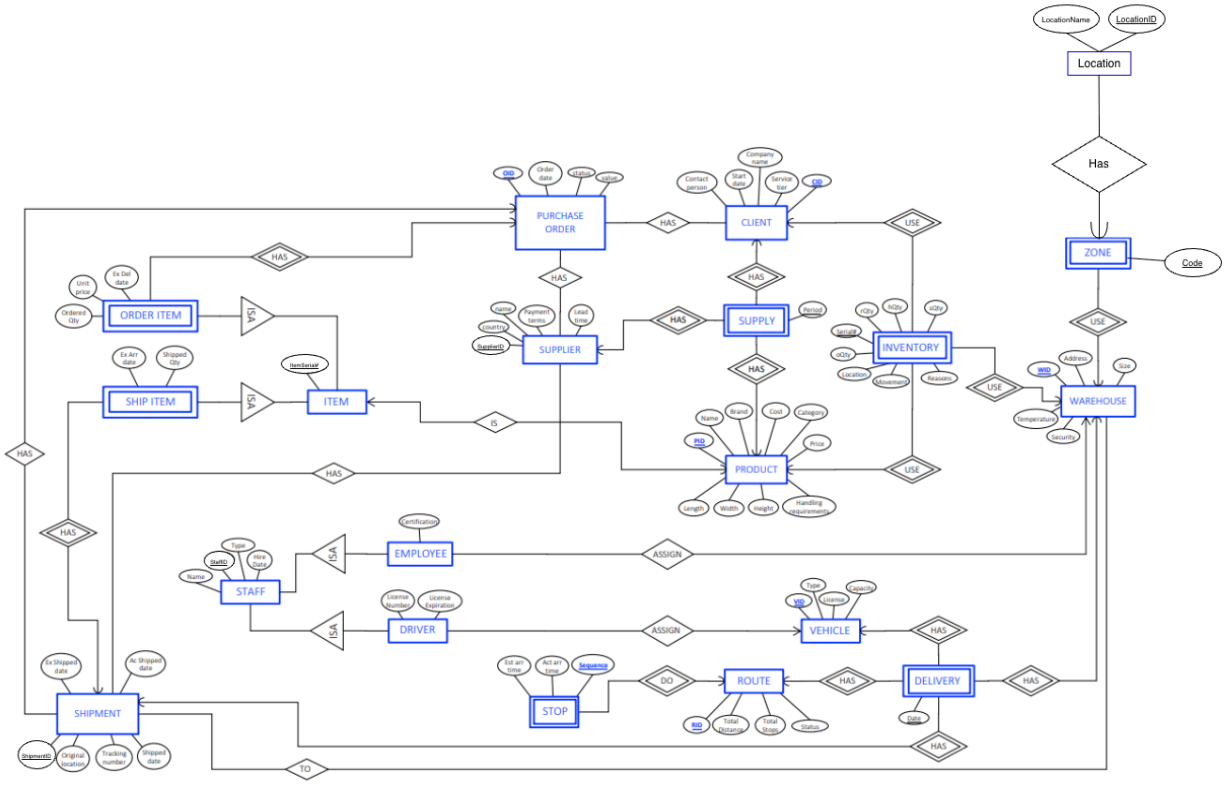
**Final ER diagram:**




**R1 - CLIENT(CID, CompanyName, ContactPerson, StartDate, ServiceTier)**

Keys: CID

Primary Key: CID

FDs:

1. CID → CompanyName, ContactPerson, StartDate, ServiceTier

The relation is in 3NF.

**R2 - SUPPLIER(SupplierID, Name, Country, PaymentTerms, LeadTime)**

Keys: SupplierID

Primary Key: SupplierID

FDs:

1. SupplierID → Name, Country, PaymentTerms, LeadTime

The relation is in 3NF.

**R3 - PRODUCT(PID, Name, Brand, Cost, Price, Category, Length, Width, Height, HandlingRequirements)**

Keys: PID

Primary Key: PID

FDs:

1. PID → Name, Brand, Cost, Price, Category, Length, Width, Height, HandlingRequirements

The relation is in 3NF.

**R4 - WAREHOUSE(WID, Address, Size, Temperature, Security)**

Keys: WID

Primary Key: WID

FDs:

1. WID → Address, Size, Temperature, Security

The relation is in 3NF.

**R5 - STAFF(StaffID, Name, HireDate, Type)**

Keys: StaffID

Primary Key: StaffID

FDs:

1. StaffID → Name, HireDate, Type

The relation is in 3NF.

**R6 - VEHICLE(VID, License, Type, Capacity)**

Keys: VID

Primary Key: VID

FDs:

1. VID → License, Type, Capacity

The relation is in 3NF.

**R7 - ROUTE(RID, Date, TotalDistance, TotalStops, Status)**

Keys: RID

Primary Key: RID

FDs:

1. RID → TotalDistance, TotalStops, Status

The relation is in 3NF.

**R8 - EMPLOYEE(StaffID, Certification, WID)**

Keys: StaffID

Primary Key: StaffID

FDs:

1. StaffID → Certification, WID

The relation is in 3NF.

**R9 - DRIVER(StaffID, LicenseNumber, LicenseExpiration, VID)**

Keys: StaffID

Primary Key: StaffID

FDs:

1. StaffID → LicenseNumber, LicenseExpiration, VID

The relation is in 3NF.

**R10 - ITEM(ItemSerial#, PID)**

Keys: ItemSerial#

Primary Key: ItemSerial#

FDs:

1. ItemSerial# → PID

The relation is in 3NF.

**R11 - ORDER_ITEM(OID, ItemSerial#, OrderedQty, UnitPrice, ExDelDate)**

Keys: (OID, ItemSerial#)

Primary Key: (OID, ItemSerial#)

FDs:

1. OID, ItemSerial# → OrderedQty, UnitPrice, ExDelDate

The relation is in 3NF.

**R12 - SHIP_ITEM(ShipmentID, ItemSerial#, ShippedQty, ExArrDate)**

Keys: (ShipmentID, ItemSerial#)

Primary Key: (ShipmentID, ItemSerial#)

FDs:

1. ShipmentID, ItemSerial# → ShippedQty, ExArrDate

The relation is in 3NF.

**R13 - ZONE(WID, Code)**

Keys: (WID, Code)

Primary Key: (WID, Code)

FDs:

1. WID, Code → WID, Code (No non trivial FDS)

The relation is in 3NF.

**R14 - LOCATION(LocationID, LocationName, WID, Code)**

Keys: LocationID

Primary Key: LocationID

FDs:

1. LocationID → LocationName, WID, Code

The relation is in 3NF.

**R15 - STOP(RID, Sequence, EstArrTime, ActArrTime)**

Keys: (RID, Sequence)

Primary Key: (RID, Sequence)

FDs:

1. (RID, Sequence) → EstArrTime, ActArrTime

The relation is in 3NF.

**R16 - DELIVERY(RID, VID, WID, ShipmentID, DeliveryDate)**

Keys: (RID, VID, WID, ShipmentID, DeliveryDate)

Primary Key: (RID, VID, WID, ShipmentID, DeliveryDate)

FDs:

1. RID, VID, WID, ShipmentID, DeliveryDate → RID, VID, WID, ShipmentID, DeliveryDate

The relation is in 3NF.

**R17 - PURCHASE_ORDER(OID, OrderDate, Status, Value)**

Keys: OID

Primary Key: OID

FDs:

1. OID → OrderDate, Status, Value

The relation is in 3NF.

**R18 - CLIENT_PURCHASE_ORDER(OID, CID)**

Keys: (OID, CID)

Primary Key: (OID, CID)

No non-trivial FDs. The relation is in 3NF.

**R19 - SUPPLIER_PURCHASE_ORDER(OID, SupplierID)**

Keys: (OID, SupplierID)

Primary: (OID, SupplierID)

No non-trivial FDs. The relation is in 3NF.

**R20 - INVENTORY(PID, WID, CID, Serial#, rQty, hQty, sQty, oQty, Movement, Reasons)**

Keys: (PID, WID, CID, Serial#)

Primary Key: (PID, WID, CID, Serial#)

FDs:

1. (PID, WID, CID, Serial#) → rQty, hQty, sQty, oQty, Movement, Reasons

The relation is in 3NF.

**R21 - SUPPLY(CID, SupplierID, PID, Period)**

Keys: CID, SupplierID, PID

Primary Key: CID, SupplierID, PID

FDs:

1. CID, SupplierID, PID → Period

The relation is in 3NF.

**R22 - SHIPMENT(ShipmentID,OriginalLocation,TrackingNumber,ShippedDate,ExShippedDate,AcShippedDate,OID)**

Keys: ShipmentID

Primary Key: ShipmentID

FDs:

1. ShipmentID → OriginalLocation, TrackingNumber, ShippedDate, ExShippedDate, AcShippedDate,OID

The relation is in 3NF.

**R23- SHIPMENT_TO_WAREHOUSE(ShipmentID, WID)**

Keys: (ShipmentID, WID)

Primary key: (ShipmentID, WID)

No non-trivial FDs. The relation is in 3NF.

# APPENDIX C: INDIVIDUAL CONTRIBUTION FORM

| Full Name | Individual Contribution to Lab 1 Submission | Percentage of Contribution | Signature |
|---|---|---|---|
| Madhav Raghu Anantharam | Write up, fact check, resolved inconsistencies | 20% | |
| Samarth Bose | Created diagram skeleton with entities, relations, attributes. Reviewed inconsistencies | 20% | |
| Serena Dai Yi | Reviewed ER diagram and writeup | 20% | |
| Clarice Kevin Stansilas | Adding attributes, diagram presentation | 20% | |
| Singh Ashna | Adding relationships to ER diagram, final checks | 20% | |
| | | | |
| | | | |

| Full Name | Individual Contribution to Lab 5 Submission | Percentage of Contribution | Signature |
|---|---|---|---|
| Clarice Kevin Stansilas | Functional dependencies, updates to ER diagram | 20 | |
| Samrath Bose | Created FD List, ER diagram, Relations | 20 | |
| Madhav Raghu Anantharam | added relations, FDs and resolved inconsistencies | 20 | |
| Serena Dai Yi | Refine FDs and ER diagram | 20 | |
| Singh Ashna | Refine FD and Relations list | 20 | |

# APPENDIX D: USE OF AI TOOL(S) IN LAB WORK

Each team member should indicate either A or B:

A. I affirm that my contribution(s) to the lab work is my own, produced without help from any AI tool(s).
B. I affirm that my contribution(s) to the lab work has been produced with the use of AI tool(s).

| Team member (full name) | Signature | Date | A or B |
|---|---|---|---|
| Madhav Raghu Anantharam | | 12/3/26 | A |
| Samarth Bose | | 12/3/26 | A |
| Serena Dai Yi | | 12/3/26 | A |
| Clarice Kevin Stansilas | | 12/3/26 | A |
| Singh Ashna | | 12/3/26 | A |
| | | | |
| | | | |

By signing this form, you declare that the above affirmation made is true and that you have read and understood NTU's policy on the use of AI tools.

If any team member answered B, the team member(s) must indicate and replicate the table below for every instance that AI tool(s) is used:

| | |
|---|---|
| Name of AI tool | < For example, ChatGPT > |
| Input prompt | < Insert the question that you asked ChatGPT > |
| Date generated | |
| Output generated | < Insert the response verbatim from ChatGPT > |
| Output screenshots | |
| Impact on submission | < Briefly explain which part of your submitted work was ChatGPT's response applied > |